Debounce Test Sheet / Test Log

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Webinar 2 — GPIO / Interrupt / Button Bouncing

Environment

Parameter	Value
Board	ESP32 DevKit v1
Input	BUTTON / NOISE → GPIO18
Output	LED → GPIO2
Simulator	Wokwi
Serial	115200
Build	GOOD / BUGGY: BUGGY

Acceptance Criteria

Valid physical presses = logical press events. RAW interrupts may be greater. Short unstable noise must not create a false logical press/action.

Test Cases

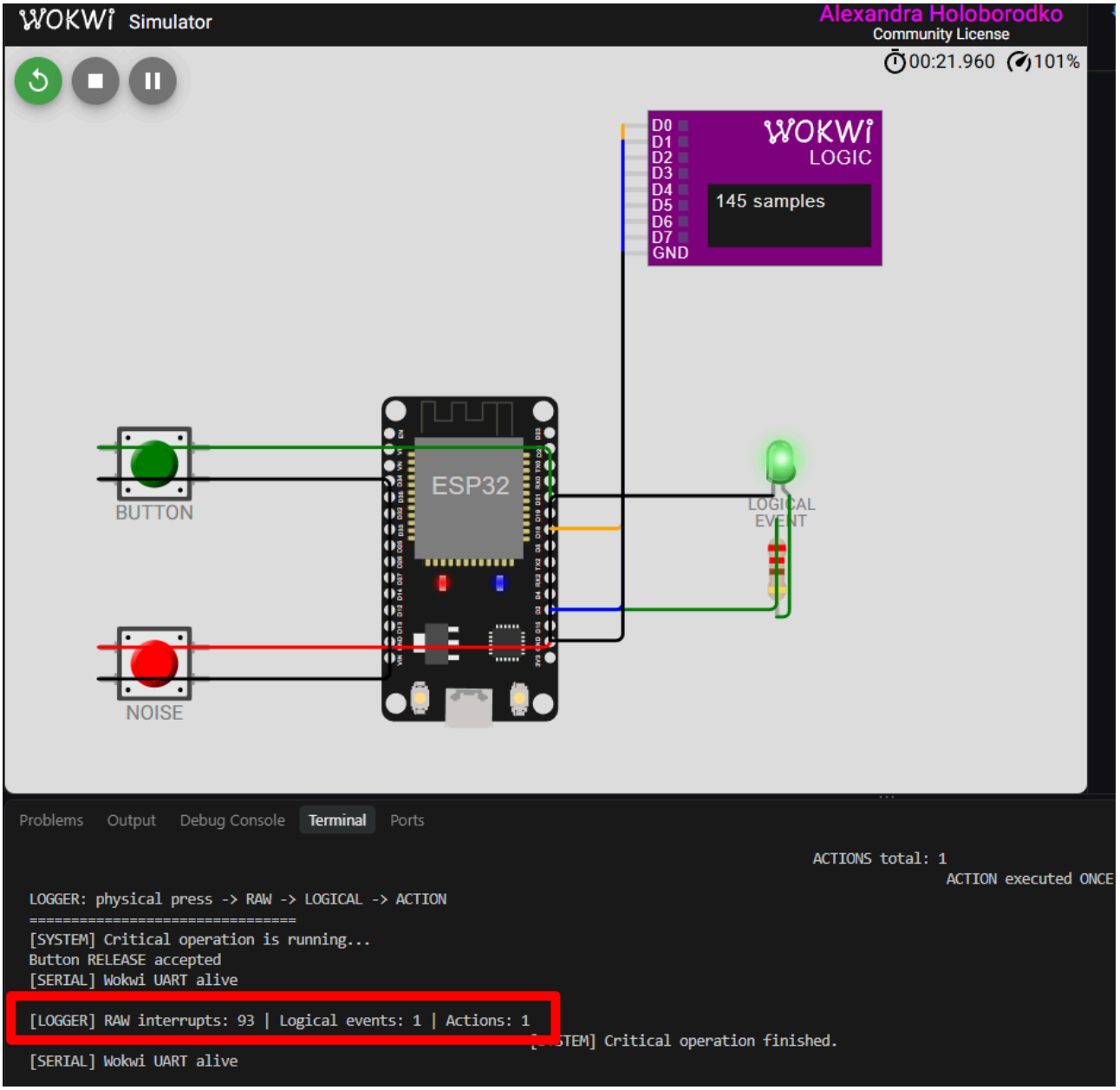
ID	Title	Steps	Expected	Actual	P/F
TC-01	Normal press	Press/release `BUTTON` once	1 physical press → 1 logical event → 1 action	1 physical press → 1 logical event → 1 action	P
TC-02	Normal press in Logic Analyzer	Observe one normal press in Logic Analyzer	D0 may contain multiple transitions; D1 changes only once for accepted press; Logger records one logical event/action	D0 (RAW)=93; D1 (LOGIC) =1; Wokwi Logic: 145	P
TC-03	Long press	Hold `BUTTON` for 2–5 s	No repeated press events/actions while held	No repeated press events/actions while held	P
TC-04	Rapid presses	Press `BUTTON` 5 separate times	5 physical presses → 5 logical press events → 5 actions	5 physical presses → 5 logical press events → 5 actions	P
TC-05	Short noise pulse	Generate a short `NOISE` pulse	RAW activity is allowed; no false logical press/action	RAW activity is allowed; false logical press/action is present	F
TC-06	Noise during critical operation	Press `BUTTON`, then generate `NOISE` during critical operation	No extra logical event/action; operation continues	extra logical event/action	F
TC-7	Most suspicious scenario	Repeat the most suspicious scenario 5 times	TC-5 & TC-6	Reproducibility is always	F

Execution Log

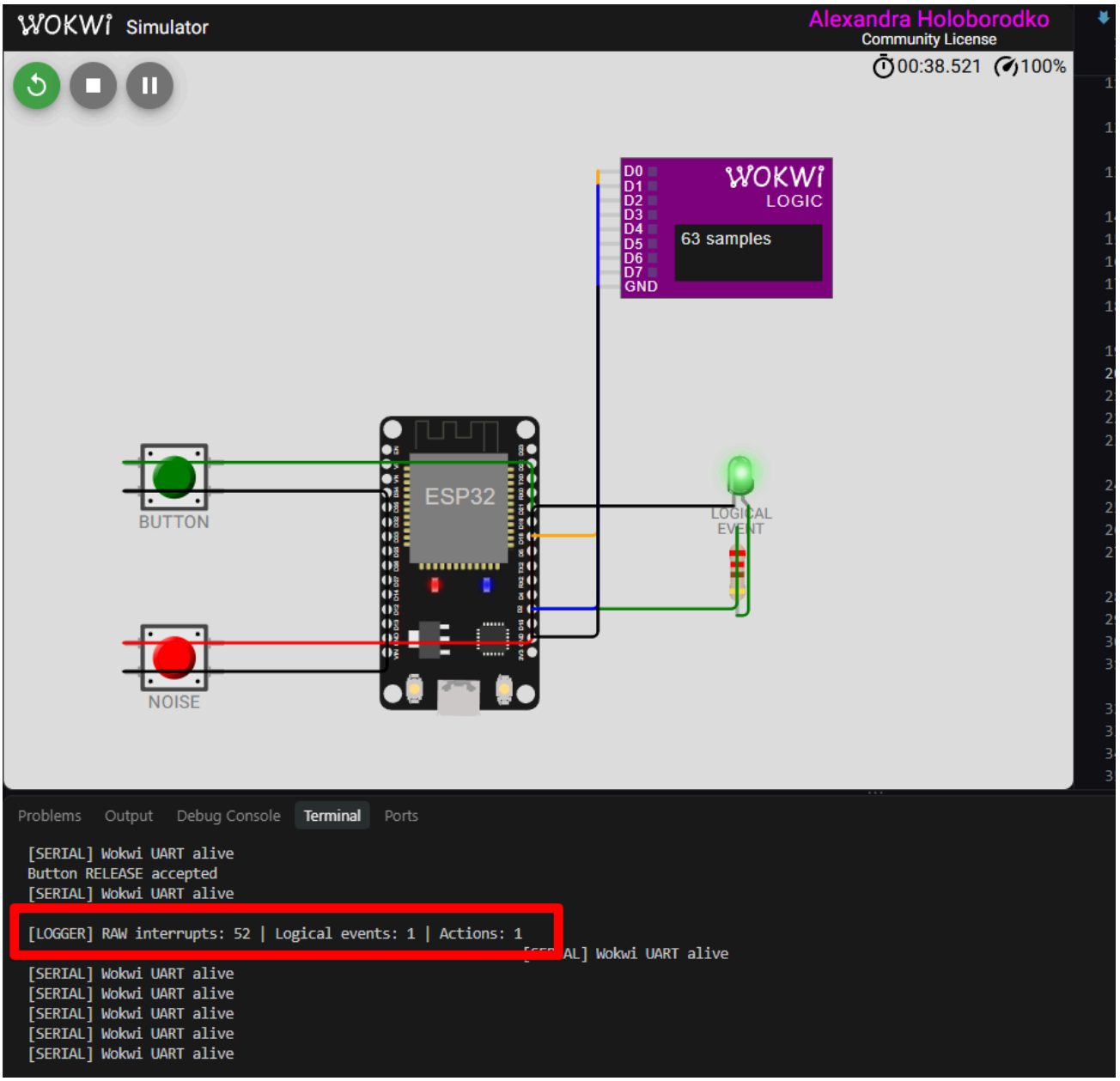
ID	Scenario	Physical presses	RAW	Logical	Actions	Expected	Actual	P/F
TC-01	Normal press	1	93	1	1	1	1	P
TC-02	Natural/Short bounce Heavy bounce	1	93	1	1	1	1	P
TC-03	Long press	1	52	1	1	1	1	P

TC-04	Rapid presses	5	LOGICAL EVENT #5: 582 (total) [LOGGER] RAW interrupt: 677 See screens below Which number should we count?	5	5	5	5	P
TC-05	Short noise pulse	1	RAW interrupts total: 1 [LOGGER] RAW interrupts: 2	1	1	1	1	F
TC-06	Noise during critical operation	2	RAW interrupts total: 175 [LOGGER] RAW interrupts: 176	2	2	2	2	F

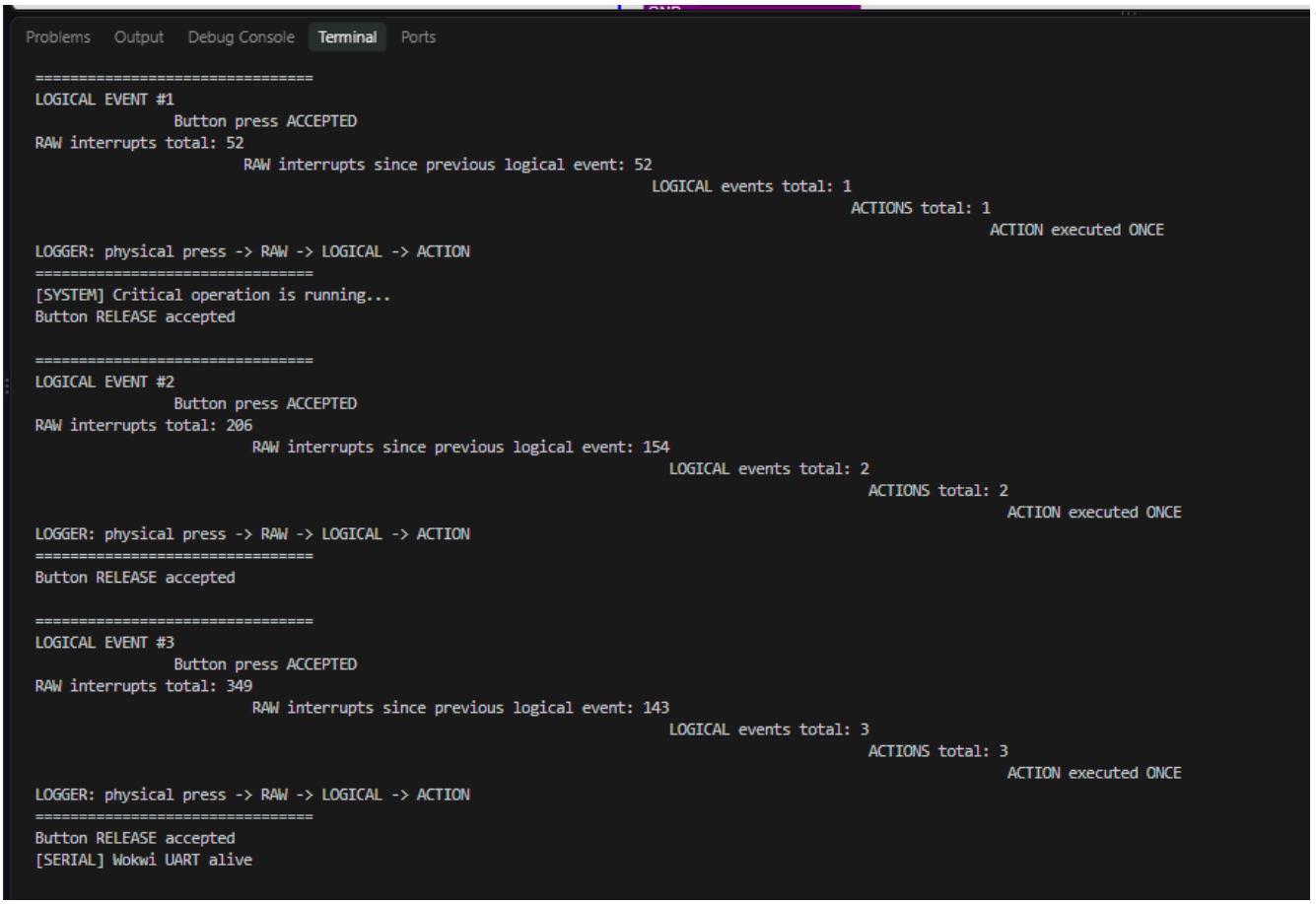
TC-01 Normal press / TC-02 Normal press in Logic Analyzer

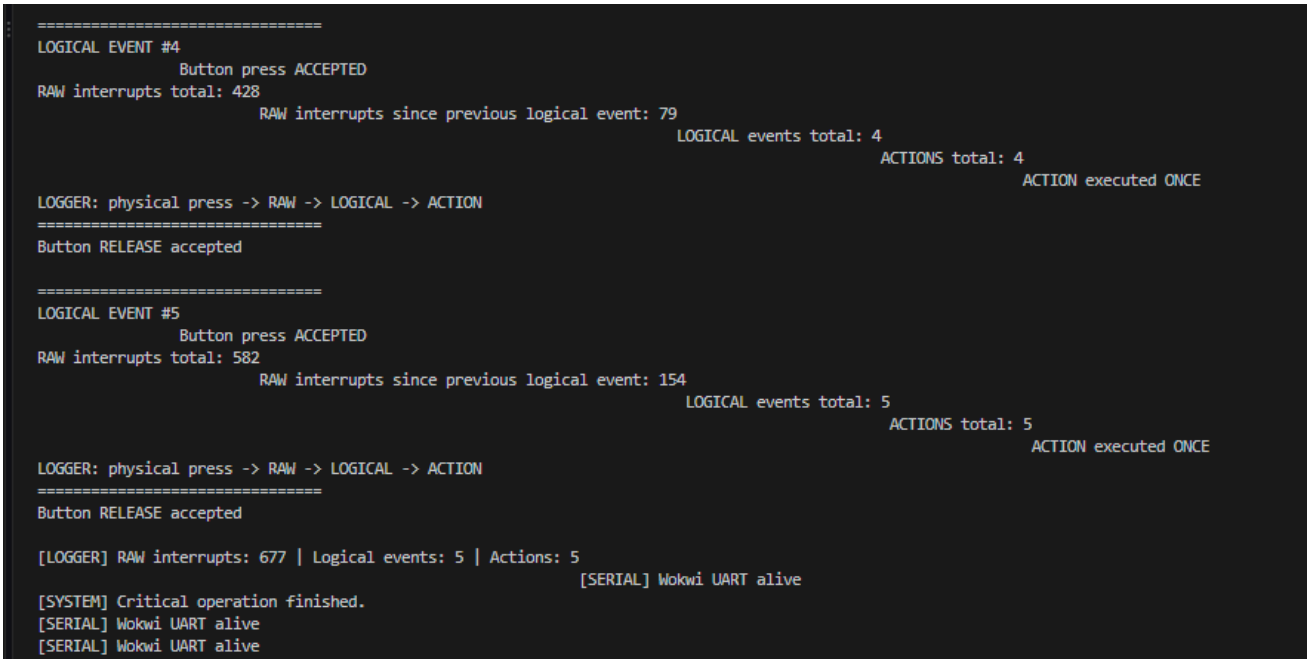


TC-3 Long press

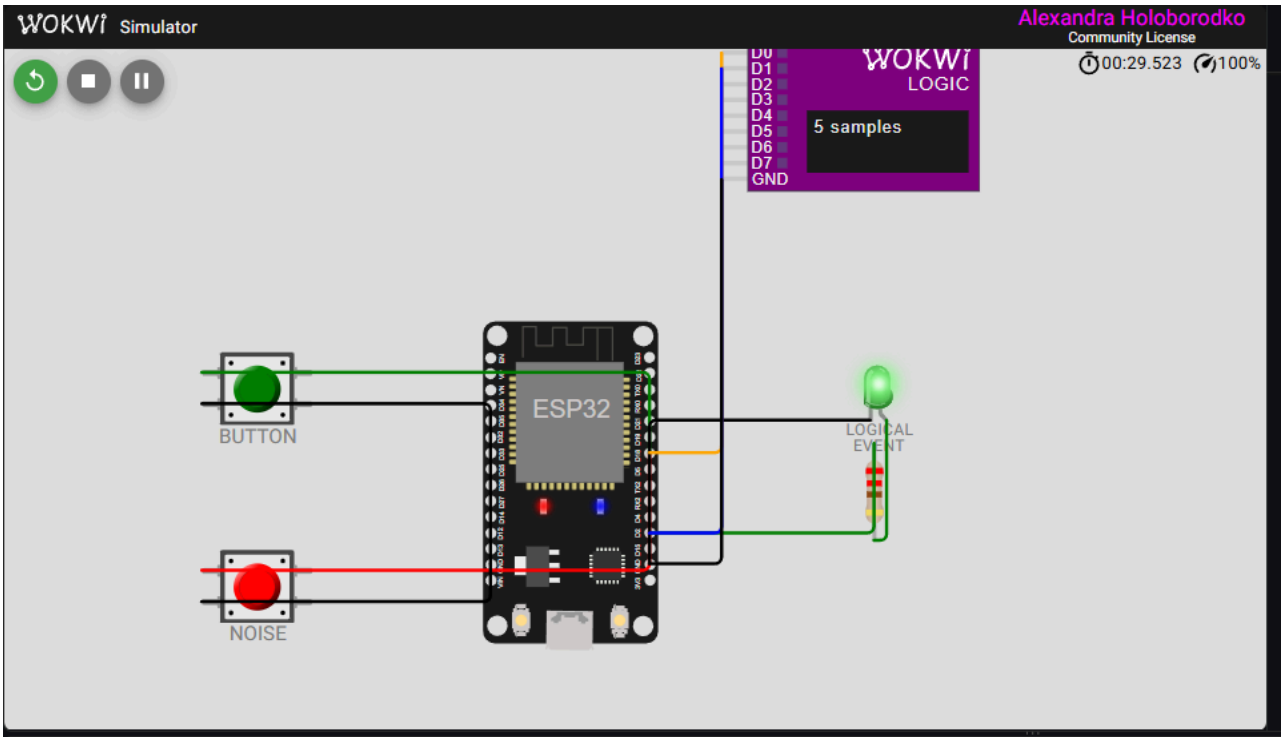


TC-4 Rapid presses





TC-5 Short noise pulse



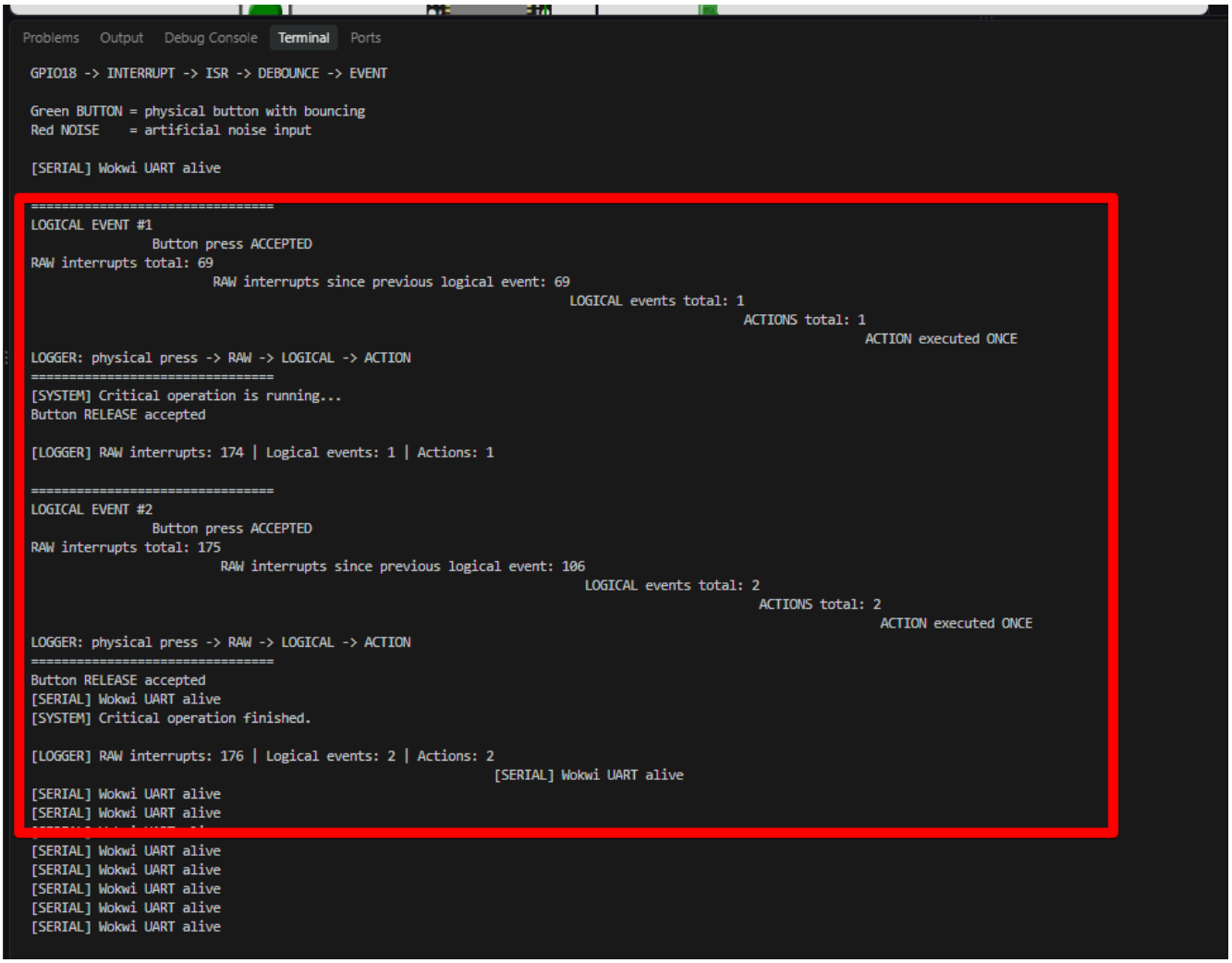

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Problems  Output  Debug Console  Terminal  Ports

[SERIAL] Wokwi UART alive

==============================
LOGICAL EVENT #1
              Button press ACCEPTED
RAW interrupts total: 1
                    RAW interrupts since previous logical event: 1
                                                          LOGICAL events total: 1
                                                                                ACTIONS total: 1
                                                                                               ACTION executed ONCE
LOGGER: physical press -> RAW -> LOGICAL -> ACTION
==============================
[SYSTEM] Critical operation is running...
Button RELEASE accepted
[SERIAL] Wokwi UART alive
[SYSTEM] Critical operation finished.
[SERIAL] Wokwi UART alive

[LOGGER] RAW interrupts: 2 | Logical events: 1 | Actions: 1
                                                    [SERIAL] Wokwi UART alive
[SERIAL] Wokwi UART alive
[SERIAL] Wokwi UART alive
[SERIAL] Wokwi UART alive
```

TC-6 Noise during critical operation



Bugs Report

Bug_1 (TC_5)

Title: Extra logical event/action is present after press NOISE button

Environment: ESP32 DevKit v1

Preconditions:simulation is started

Steps to Reproduce:

1. Press NOISE button
2. Release NOISE button
3. Check Serial Monitor

Expected Result: no false logical press/action

Actual Result: logical press/action is present after press NOISE button

Evidence: TC-5 Short noise pulse screen above

Reproducibility: always

Severity: Critical

Bug_2 (TC_6)

Title: Extra logical event/action is present after press NOISE button during critical operation

Environment: ESP32 DevKit v1

Preconditions: simulation is started

Steps to Reproduce:

1. Press “Button” button
2. Release “Button” button
3. Press “NOISE” button during critical operation
4. Release “NOISE” button
5. Check Serial Monitor

Expected Result: no false logical press/action

Actual Result: logical press/action is present after press NOISE button

Evidence: TC-6 Noise during critical operation screen above

Reproducibility: always

Severity: Critical

QA Conclusion

Debounce is not reliable because during testing 2 critical issues were found - Noise button press provokes logical event and action which is contradicts AC.

Risk: system works in invalid way (?)

Additional checks: ? need help with it

Bouncing on press/release? But is it even possible to reproduce this in the simulator?